Department of Electrical Engineering
University of Arkansas

ELEG 3143 Probability & Stochastic Process

Ch. 0 Preface

Dr. Jingxian Wu
wuj@uark.edu

GENERAL INFORMATION

- **Instructor: Dr. Jingxian Wu**
  - Email: wuj@uark.edu
  - Phone: (479) 575-6584
  - Office Bell 3168

- **Office Hours**
  - Tu. Th. 1:00 PM – 2:00 PM,
  - By appointment

- **Lecture Schedule**
  - Bell 2273
  - Tu. Th. 11:00 AM – 12:15 PM

TEXTBOOK AND REFERENCES

- **Required Text Books**
  - R. D. Yates and D. Goodman, *Probability and Stochastic Processes: A Friendly Introduction for Electrical and Computer Engineers*, 3rd Edition, Wiley, May 2014.

- **Required Software**
  - Matlab

- **References (optional)**
  - S. M. Ross, *Introduction to Probability Models*, 9th Ed., Academic Press, 2007.
  - A. Papoulis and S. U. Pillai, *Probability, Random Variables and Stochastic Processes*, 4th Ed., McGraw Hill, 2002.
  - G. R. Cooper and C. D. McGillem, *Probabilistic Methods of Signal and System Analysis*, 3rd Ed., Oxford University Press, 1999.

COURSE INFORMATION

- **Pre-requisite:**
  - Signal and System
  - Calculus I, II, & III

- **This course involves heavy mathematical derivations**
  - It is a Mathematics course.
  - A large number of examples will be given in class – It's ESSENTIAL for you to repeat all the examples by yourself after class.

- **Teaching format**
  - Slides
  - Examples
  - Exercises
  - Homework assignments (problems, software)

TENTATIVE SCHEDULE

- Week 1 (1/15, 1/17): Ch.1 Introduction to Probability
- Week 2 (1/22, 1/24): Ch. 1 Introduction to Probability
- Week 3 (1/29, 1/31): Ch. 1 Introduction to Probability
- Week 4 (2/5, 2/7): Ch. 2 Discrete Random Variables
- Week 5 (2/12, 2/14): Ch. 2 Discrete Random Variables
- Week 6 (2/19, 2/21): Ch. 3 Continuous Random Variables (Test 1 on 2/21)
- Week 7 (2/26, 2/28): Ch. 3 Continuous Random Variables
- Week 8 (3/5, 3/7): Ch. 4 Multiple Random Variables
- Week 9 (3/12, 3/14): Ch. 4 Multiple Random Variables
- Week 10 (3/19, 3/21): Spring Break
- Week 11 (3/26, 3/28): Ch. 4 Multiple Random Variables
- Week 12 (4/2, 4/4): Ch. 5 Elements of Statistics (Test 2 on 4/4)
- Week 13 (4/9, 4/11): Ch. 5 Elements of Statistics
- Week 14 (4/16, 4/18): Ch. 6 Stochastic Process
- Week 15 (4/23, 4/25): Ch. 6 Stochastic Process
- Week 16 (4/30, 5/2): Ch. 6 Stochastic Process (dead day: 5/3)
- Test 3 at the Final week

GRADING POLICY

- **Grades Percentage**
  - Test 1 23%
  - Test 2 23%
  - Test 3 23%
  - Homework 23%
  - Quiz 8%

- **Grades**
  - A: $90 \leq \text{grade} \leq 100$
  - B: $80 \leq \text{grade} < 90$
  - C: $70 \leq \text{grade} < 80$
  - D: $60 \leq \text{grade} < 70$
  - F: $0 \leq \text{grade} < 60$

GRADING POLICY

- **All homework need to be directly uploaded to blackboard before the due date.**
- **Due dates for homework and lab report will be strictly enforced. Late submission within one week after due will receive a 20% deduction; no credit if submitted one week past due.**
- **There will be NO make up for quizzes.**
- **If for some legitimate reason (sickness, death in the family, etc.), you cannot take an exam on the scheduled day, you must notify the instructor prior to the exam.**

ONLINE RESOURCES

- **Course Home Page**
  - https://wuj.hosted.uark.edu/teaching/eleg3143/eleg3143.html
  - Blackboard
  - All the course related materials, such as slides, homework assignments, links, announcements, etc., will be posted on this website.
  - Please check the webpage regularly (at least once per week) for update.

ADDITIONAL ISSUES

- **Academic Honesty**
  - Each University of Arkansas student is required to be familiar with and abide by the University's 'Academic Integrity Policy' which may be found at http://provost.uark.edu/
  - Any kind of activities related to academic dishonesty (copying homework, lab report, code, plagiarism, etc.) will be dealt with.
  - If you are not sure about plagiarism, please contact the instructor.
- **Questions are welcome in my class**
  - You are very welcome to raise any question related to course materials.
  - Please feel free to stop me at any time if you have any question.
  - You can also ask me question via email or during office hours.
- **To respect your fellow students as well as the instructor, please turn off or silencing your cell phone.**
  - No text messaging or web surfing!
- **BE ON TIME!**
- **Have Fun!**